

SPE 77399

Quantification of Multi-Phase Fluid Saturations in Complex Pore Geometries From Simulations of Nuclear Magnetic Resonance Measurements

E. Toumelin, SPE, C. Torres-Verdín, SPE, The University of Texas at Austin, and S. Chen, SPE, Baker Atlas

Copyright 2002, Society of Petroleum Engineers Inc.

This paper was prepared for presentation at the SPE Annual Technical Conference and Exhibition held in San Antonio, Texas, 29 September–2 October 2002.

This paper was selected for presentation by an SPE Program Committee following review of information contained in an abstract submitted by the author(s). Contents of the paper, as presented, have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material, as presented, does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Papers presented at SPE meetings are subject to publication review by Editorial Committees of the Society of Petroleum Engineers. Electronic reproduction, distribution, or storage of any part of this paper for commercial purposes without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of where and by whom the paper was presented. Write Librarian, SPE, P.O. Box 833836, Richardson, TX 75083-3836, U.S.A., fax 01-972-952-9435.

Abstract

We develop a numerical algorithm to simulate nuclear magnetic resonance (NMR) measurements in the presence of constant magnetic field gradients. The algorithm is based on Monte Carlo conditional random walks in restricted and unrestricted space. Simulations can be performed of three-dimensional (3D) porous media that include both arbitrary bimodal pore distributions and multi-phase fluid saturations. The ability to account for the presence of a constant external magnetic field gradient allows us to replicate actual well logging conditions that include the effect of CMPG pulse sequences at a microscopic level. This is accomplished by simulating pulse acquisition techniques that include multiple inter-echo times (TE) similar to those currently used by the well-logging industry. Benchmark examples are presented to validate the accuracy and internal consistency of our algorithm against previously published results for the case of a null magnetic field gradient. Validation examples are also presented against actual NMR measurements performed on core samples of carbonate rock formations.

Interpretation work is focused to the petrophysical assessment of both partial oil/water saturations and pore structures exhibiting hydraulic coupling. Simulation examples are designed to quantify whether the inclusion of diffusion under a magnetic field gradient can improve the interpretation of multi-phase fluid saturations when hydraulic coupling is significant. The simulation algorithm sheds light to new NMR data acquisition strategies that could be used to improve the detection and quantification of (a) fluid types, (b) complex fluid saturations, and (c) complex pore geometries.

Introduction

Presence of hydraulic (or diffusion) pore coupling in carbonate rocks (mainly grainstones) challenges conventional NMR interpretation techniques. Although pore coupling phenomena are commonplace in the majority of rock pore systems, they become relevant to assess NMR measurements when the following three conditions are met: (a) micro-porosity regions are present within the grains, (b) micro-pores are hydraulically well connected (not cemented) to outer macro-porous regions exhibiting low surface-to-volume ratios, and (c) rock surface relaxivity is low enough to prevent decay of proton magnetization within the macro-porosity before protons can enter the micro-porous regions. In addition, fluid diffusivity must be sufficiently large in order for hydraulic coupling to be significant within the time scale of NMR measurements. This is usually the case only for water and light hydrocarbons. As a result, fluid magnetization will be exchanged between micro- and macro-pore regions, and no obvious relationship will exist between NMR transverse relaxation (T_2) distribution and pore-size distribution. **Figure 1** illustrates conditions (a) and (b) described above with an example of scanning electron microscope (SEM) images of a carbonate rock exhibiting hydraulic coupling.

Varying inter-echo times (TE) is a common NMR data acquisition technique used for *in-situ* reservoir fluid identification^{1,2,3}. However, in the case of NMR measurements performed in carbonate rocks, there exist no published reports dealing with the impact of diffusion coupling on hydrocarbon typing and quantification using multiple- TE logging techniques. The objectives of this paper are twofold: (a) to develop a simulation algorithm capable of reproducing NMR measurements in complex pore geometries under a variety of experimental conditions and, in particular, under the influence of a constant magnetic field gradient, and (b) to provide simulation examples that will help assess the validity of fluid phase discrimination using multi- TE measurements in porous media exhibiting hydraulic coupling.

In the past, numerical models of NMR decay have been proposed based on periodic bimodal packs of spheres that accounted for surface relaxation effects. We have reproduced and extended Ramakrishnan *et al.*⁴'s Monte Carlo algorithm to account for the effect of an external constant magnetic field

gradient of the type enforced by modern NMR tools, and in the presence of a non-wetting phase. The first part of this paper introduces the Monte Carlo simulation algorithm applicable to a bimodal pack of spheres in the presence of a constant magnetic field gradient. A subsequent section describes examples of numerical simulation that illustrate the versatility of the algorithm. Finally, we derive and interpret simulation examples intended to address specific issues of fluid discrimination using multiple inter-echo times in the presence of diffusion coupling. We address four specific cases of NMR T_2 distributions (two unimodal and two bimodal) by modeling several possible combinations of pore structure, hydraulic coupling, and fluid distribution. These case studies were designed such that different pore configurations created identical NMR signal at low values of TE , but did exhibit differences at high values of TE .

Model for the Simulation of NMR Decay

The algorithm developed to simulate numerically NMR magnetization decay in carbonate rocks makes use of conditional Monte Carlo random walks. It is based on the algorithm presented by Ramakrishnan *et al.*⁴, further generalized to include a microscopic description of diffusivity effects in the presence of a constant magnetic field gradient⁵. The porous medium can either include micro- and macro-porous regions (grainstone model) to form a bimodal pore distribution, or can exhibit a single pore size with solid grains (wackestone model).

Model geometry. The simulation model was constructed with a three-dimensional (3D) bimodal pack of spheres that replicates the complex pore-structure geometry encountered in some carbonate formations, namely micro-porous grainstones, where the solid micro-grains are arranged into packs. As in the example shown in **Fig. 1**, the model adopted for the simulations accounts for a micro-porosity region immediately surrounding the grains, while different types of macro-porosities (characterized by different sizes and degrees of cementation, depending on diagenesis) exist between the packs. The geometrical construction of the 3D model of a bimodal pack of spheres is based on the following assumptions:

- Bimodal pore-size distribution,
- Periodic geometry,
- Isotropic pores,
- Spherical, compacted micro-grains, and
- Spherical, compacted grain packs.

At the micro- (respectively, macro-) scale, each grain (respectively, grain pack) of the model is defined as a sphere inscribed into a concentric cubic cell. If the sphere does not completely fill its cubic unit, then the complementary volume is considered filled with fluids. Furthermore, fixed blobs centered in the macro-pore space can be included in the model to represent partial saturations of immiscible non-wetting oil phase, while the micro-porosity can only be filled with irreducible water. **Figure 2** shows a 3D period of the assumed bimodal pore structure. A large variety of pore sizes can be

modeled by adjusting the values of the grain (or grain pack) radii, and the sizes of the corresponding cubic cells.

Unimodal pore-size distributions can be constructed with the same basic model by allowing no micro-porosity. The NMR response of uncoupled bimodal pore structures, i.e., porous media where no hydraulic coupling occurs, can be simulated by superimposing the response of each isolated pore mode.

Dual random-walk strategy. At each iteration of the NMR numerical simulation algorithm, a fictitious proton is randomly chosen in the available pore space. This proton is thereafter displaced to a new location yielded by a continuous random walk operating according to two possible strategies. Depending on the proximity of the proton to the surface of material discontinuity (pore wall or fluid phase interface), either a conventional random-walk strategy, or Zheng and Chiew's First-Passage-Time technique⁶ can be implemented to displace the proton to its new location.

The First-Passage-Time strategy was developed to allow macroscopic jumps in the bulk pore space and hence to speed up the otherwise prohibitively long random-walk process. It is based on the analytical solution of the unbounded diffusivity equation, which yields a stochastic relationship between the mean square displacement, $\langle R^2 \rangle$, during a given step, and its duration, Δt , with a maximum of probability close to the free space bulk diffusivity, i.e.,

$$D_o = \frac{\langle R^2 \rangle}{6\Delta t}. \quad (1)$$

Figure 3 illustrates the selection of the random walk strategy according to the distance between the particles and the surrounding fluid boundaries. Grain surface and fluid interfaces constitute the wetting phase boundaries, whereas only the interface between fluids constitutes a non-wetting phase boundary).

Model of diffusive relaxation. In general, the apparent transversal relaxation time T_2 of the decay of magnetization signal can be written as:

$$\frac{1}{T_2} = \frac{1}{T_{2S}} + \frac{1}{T_{2B}} + \frac{1}{T_{2D}}, \quad (2)$$

where T_{2S} is the surface relaxation, T_{2B} the bulk relaxation, and T_{2D} the decay rate due to diffusion in the presence of a gradient field (diffusive relaxation). The Monte Carlo algorithm described by Ramakrishnan *et al.*⁴ takes into account the effect of surface relaxation, but does not include the effect of bulk or diffusive relaxations. On the one hand, T_{2B} is governed by fluid properties, regardless of time or geometry, and therefore can be applied independently from the Monte Carlo simulation process. On the other hand, T_{2D} depends on both NMR experimental parameters (i.e., inter-echo time TE and gradient strength G), and the effective fluid diffusivity D , as suggested by the macroscopic expression

$$T_{2D} = \left(\frac{\gamma^2 G^2 TE^2 D}{12} \right)^{-1}, \quad (3)$$

where γ is the proton gyromagnetic ratio. When molecules diffuse through restricted geometries, the otherwise free diffusion is hindered and this causes the observed apparent diffusivity to become time and geometry dependent⁷. The effective diffusivity D decays from the bulk (free space) value D_o at a rate which depends on tortuosity, time, and pore surface-to-volume ratio. Within micro-porous regions in the micrometer range, for instance, the decay of D is instantaneous to reach an asymptotic value equal to the inverse of the tortuosity of the medium. By considering the diversity of pore sizes within carbonate rocks, it then becomes necessary to track the diffusivity effect through the random walk in order to model diffusive relaxation.

Since the magnetization signal acquired by the NMR tool is proportional to the cosine of the spin phases of the fluid protons, we designed the algorithm such that it could track the spin dephasing caused by the constant magnetic field gradient. At time t , a proton spin phase is given by the integral

$$\phi(t) = \int_0^t \gamma \mathbf{G} \cdot \mathbf{r}(t') dt' = \gamma G \int_0^t z(t') dt', \quad (4)$$

where $\mathbf{r}$ is the position of the proton, and the gradient vector $\mathbf{G}$ is assumed to point in the z -direction. Due to the stochastic nature of the displacement z while using the First-Passage-Time technique, the evaluation of the integral of equation (4) is not trivial. To circumvent this problem, we made use of the phase-shift expression derived by Gudgertsson and Patz⁸, which is based on a stochastic formulation of the phase shift created by the gradient. This phase-tracking strategy was designed to honor the CPMG sequence with a sampling rate TE , as described below.

Simulation algorithm. The following algorithmic sequence was applied for each proton and at each random-walk step, i , to account for the selection of a displacement strategy and of a CPMG sampling protocol:

1. Determine the distance d to the closest material interface
2. If d is large, then the proton movement is inscribed in free space and the First-Passage-Time technique is applied^{4,6}. Due to the constraint of the TE sequence, the duration Δt is such that it cannot extend beyond a half- TE period.
3. If d is small, then the above algorithm does not converge. Therefore, it must be substituted by a conventional random-walk strategy with fixed infinitesimal displacement ϵ .
4. Regardless of d , the phase shift acquired during this step is given by⁸:

$$\phi(t_{i+1}) = \phi(t_i) + \gamma G \left(\frac{(z(t_i) - z(0)) + (z(t_{i+1}) - z(0))}{2} \right) (t_{i+1} - t_i) + \gamma G \sqrt{\frac{D_o (t_{i+1} - t_i)^3}{6}} \text{Normal}(), \quad (5)$$

where t_i , t_{i+1} and 0 are the initial instant of the step, the final instant of the step, and the time origin, respectively. In equation (5), Normal() is a Gaussian probability density function of zero mean and unity variance.

5. When $t_{i+1} = nTE$, where n is an integer, the phase cosine (i.e., the magnetization of the particle at time nTE) is recorded and added to the sum of signals of all the protons simulated at time nTE .
6. When $t_{i+1} = (n + \frac{1}{2})TE$, where n is an integer, the phase is reversed to honor the RF (π)_y re-polarizing CPMG magnetic pulses.
7. If contact is made with a surface of material discontinuity (either grain or interface between immiscible fluids), then the proton magnetization decays with a probability equal to⁹

$$p = \frac{\rho \epsilon}{D_o}, \quad (6)$$

where ρ is the surface relaxivity of the interface. If no magnetization decay occurs, then the proton rebounds specularly at the surface of the interface.

Once a large-enough number of proton trajectories are simulated with the Monte Carlo algorithm, convergence to the average time solution is insured and a total magnetization signal is output. Generally, a few hundreds to a thousand proton trajectories are needed for convergence, depending on the modeled pore-size distribution. The magnetization signal is subsequently inverted into a transverse relaxation (T_2) distribution by means of a curvature-smoothing regularization¹⁰ generally used to invert noisy experimental NMR magnetization decays. **Figure 4** is a flowchart that graphically describes the details of the Monte-Carlo algorithm.

Testing and Generalization of the Algorithm.

Effect of surface relaxivity on hydraulic pore coupling in the absence of a magnetic field gradient. In order to establish a reference for non-gradient experiments, we first conducted NMR simulations to illustrate the effect of surface relaxivity ρ on diffusion coupling. An inter-echo time $TE=1$ ms was adopted in the presence of a null magnetic field gradient. **Figure 5** shows an excellent agreement between our simulation results and those published by Ramakrishnan *et al.*⁴ for similar geometrical and fluid properties. A limitation exists to the accuracy of our simulations for small TE sampling in the case of large values of ρ . The principle of hydraulic pore coupling is brought to light by considering the spatial molecular dispersion illustrated of **Fig. 6**. Hydraulic pore

coupling increases when ρ decreases: the smaller the value of ρ , the higher the proton exchange, and therefore the higher the magnetization exchange between pores.

Effect of dual fluid saturation on the NMR response of coupled pore modes. When immiscible oil blobs are included in the model, the amount of wetting water available for coupled magnetization is naturally reduced. Therefore, the less the water saturation (S_w), the less the transfer of magnetization between the two pore scales, thereby rendering diffusion coupling negligible. This phenomenon is illustrated in **Fig. 7**. Again, our simulation results are in very good agreement with those published by Ramakrishnan *et al.*⁴

Comparison with analytical relaxation in free diffusion. To test the consistency and accuracy of the dephasing and phase-reversing strategies included in our NMR simulation algorithm, an unbounded medium was simulated with grain diameters tending to zero for both 1-cp and 6-cp oils, at a temperature of 370 K. Results from this simulation exercise are shown in **Fig. 8**. An almost perfect agreement is found between the macroscopic analytical expression given in equation (4), and the microscopic numerical results provided by our Monte Carlo simulation algorithm. Once bulk relaxation is applied to the results of the simulations, the match with analytical values, i.e., $1/(1/T_{2B} + 1/T_{2D})$, is even closer.

Example of simulations of NMR measurements acquired in a coupled carbonate rock. Under the assumption of hydraulic coupling between existing pore modes, we reached very good agreements between experimental and simulation results. For instance, we were able to consistently calibrate the sphere-pack simulation model to geometrical pore core data¹¹. **Figure 9** provides an example of such simulations results. The numerical simulations compare very well to experimental NMR measurements, including the case in which a magnetic field gradient is present. Based on the temperature dependence of the parameters D_o and T_{2B} used in the simulation algorithm, we were able to assess the relative effects of diffusion coupling and temperature on NMR measurements¹¹.

Application of the Simulation Algorithm to Fluid Discrimination with Multiple TE Measurements.

NMR T_2 distributions acquired in carbonate formations often exhibit either a unimodal or a bimodal pattern. Other than the trivial case in which the saturating fluid is irreducible water, the interpretation of unimodal and bimodal distributions is not straightforward in the presence of non-negligible pore-coupling effects. Further, when both wetting (water) and non-wetting (oil) phases are present, the overall T_2 distribution is collectively influenced by all of these mechanisms. Previous work on using multiple- TE logging measurements to identify oil saturation and viscosities^{1,2,3} was entirely based on the assumption of uncoupled systems. Several NMR simulation examples were designed in oil/water systems to assess pore-

coupling effects on multiple- TE NMR data acquisition techniques used for hydrocarbon typing.

We consider four central case studies for the simulation/interpretation studies reported in this paper. The four cases are classified according to the T_2 distribution pattern measured at inter-echo echo times of 0.25 ms, and assuming a magnetic field gradient strength of 15 G/cm. Simulation results are obtained for inter-echo times of 1 and 4 ms. According to the formulated hypotheses described below, the models used in the simulations account for either a single pore size (for unimodal and uncoupled bimodal structures) or else two coupled pore sizes (for coupled bimodal structures). The nominal pore sizes, as defined on the corresponding pore-size distributions of **Fig. 10** through **13**, were obtained by adjusting the geometrical dimensions of the unit cells of the periodic models. In all cases, the ratio between grain diameter and concentric cube side was kept constant and equal to 1.23, the surface relaxivity at the interface between fluids was neglected, and the grain surface relaxivity was held equal to 3 $\mu\text{m/s}$. Both light (1.5 cp) and medium (6 cp) oil grades were considered as non-wetting fluids, whose bulk properties are summarized in **Table 1**.

Case 1: Single T_2 peak at high T_2 values (1000 ms). (Fig. 10)

Hypothesis 1 (C1H1): Unimodal pore size distribution only filled with water. The T_2 peak shifts to the left due to the increased role played by T_{2D} on the NMR time relaxation for $TE=4$ ms.

Hypothesis 2 (C1H2): Same as C1H1, but including blobs of light oil in the pores. At low TE values, the light-oil bulk relaxation equals the surface relaxation of the water filling the rest of the pore space. In comparison with C1H1, this oil grade decays slower than water with increasing TE . However, the difference in total signal is not noticeable.

Case 2: Single T_2 peak at average T_2 values (200 ms). (Fig. 11)

Hypothesis 1 (C2H1): Unimodal pore size distribution filled with only water. The nominal pore size is smaller than in C1H1.

Hypothesis 2 (C2H2): Comparable to C1H2, but with a medium-grade oil and a slightly smaller pore size than in C1H2. Oil bulk relaxation is assumed equal to water surface relaxation within the complementary pore space. The dominant oil saturation dictates the NMR behavior of the entire system. Since the bulk relaxation of the prominent oil saturation has a magnitude similar to its diffusive relaxation in bulk space (i.e., $D \sim D_o$), the increment in echo time only slightly modifies its T_2 peak value. We remark that the total black curve shown in the corresponding panel of **Fig. 12** represents the joint response due to both fluids, and masks the NMR response of water peaking at $T_2=100$ ms (as in the case C2H1).

Hypothesis 3 (C2H3): Bimodal, coupled pore structure with micro- and macro-porosity regions, containing water

only. This pore system behaves in similar fashion to the unimodal C2H1 case described above, except for the case $TE=4$ ms in which the T_2 peak's relaxation time is equal to half of that obtained for C2H1 (55 ms vs. 100 ms). This means that hydraulically coupled pores manifest themselves as longer diffusion lengths, whereupon their magnetization signal decays faster than the equivalent signal due to hydraulically uncoupled systems. In hydraulically uncoupled systems, the macro-pores maintain the total T_2 response at values higher than that of micro-pores, whereas in hydraulically coupled systems the contribution due to micro-pore dominates the diffusive process thereby leading to lower decay times.

Hypothesis 4 (C2H4): Bimodal, coupled pore structure including medium-grade oil in the macro-porous regions. The oil bulk relaxation is close to that of the coupled response described for C2H3. The presence of medium oil maintains the relaxation value of the T_2 peak of the entire system higher than its value in C2H3, hence equal to that of the unimodal pores of case C2H1. Since oil occupies part of the macro-pore space, less space is available for water, whereupon hydraulic coupling becomes weak. On the one hand, presence of residual water causes lower decay times because the proportion of micro-porous borne water is higher in the coupling mode. On the other hand, the oil phase exhibits a T_2 shift much lower than water when TE increases. Collectively, these two adverse effects make the T_2 response of the system just slightly more spread than that of case C2H1.

Case 3: Double T_2 peak at low and high values of T_2 (50 and 1000 ms). (Fig. 12) As mentioned previously, the short T_2 peak can be solely due to irreducible fluid. Hence, this configuration can only be representative of uncoupled pore systems, or else coupled systems that remain effectively uncoupled by the complete oil saturation of the macro-pores. Therefore, this case is more typical of water-oil mixtures.

Hypothesis 1 (C3H1): Bimodal uncoupled pore size distribution, filled with water only. The two distinct T_2 peaks originating from the pore size segregation converge into a single peak when T_{2D} dominates the NMR response of the system.

Hypothesis 2 (C3H2): Bimodal uncoupled pore structure with partial saturation of medium oil. Due to the contrast of relaxation times of the two fluids, for $TE=0.25$ ms, between the oil peak (in red) and the water peaks (cf. C3H1), the response of the entire system causes a shift of the T_2 value of its macro-porosity peak (500 ms vs. 1000 ms for full water saturation). When TE increases, this shift is reinforced and splits the magnetization between two modes for $TE=4$ ms.

Hypothesis 3 (C3H3): Same configuration as in C3H2, but with light oil. The fact that light-oil bulk relaxation is closer to 1000 ms than that of medium oil causes the NMR response at low echo time to be the same as that for full water saturation (C3H1). However, a large T_2 spread appears at the longest echo times where, on the contrary, water alone causes a sharp, selective T_2 decay. The difference between this and the medium oil case C3H2 is very subtle and requires further inspection.

Hypothesis 4 (C3H4): Bimodal coupled or uncoupled pore structure, with complete light-oil saturation of the macro-pores. This case is mainly characterized by a complete segregation of the oil and water T_2 peaks at large echo times. Given the important size contrast between micro- and macro-pores (a factor of 40), such a situation provides obvious evidence of the complete saturation of the macro pores with light oil (see also C4H2). Case C3H4 represents a good-quality oil reservoir with no free water, in which case diffusive coupling has a negligible impact on NMR interpretation.

Case 4: Double T_2 peak at average and high values of T_2 (100 and 1000 ms). (Fig. 13)

Hypothesis 1 (C4H1): Bimodal uncoupled system, same as C3H1 but with larger micro-pores (which become meso pores).

Hypothesis 2 (C4H2): Same configuration as C3H4, with light oil and meso-pores instead of micro-pores. The ratio of macro- to meso-pore size is smaller (i.e., 15) than the ratio of macro- to micro-pore size of case C3H4 (i.e., 40). This difference is enough to explain the fact that in case C3H4 two distinct peaks appear for $TE=4$ ms, whereas in case C4H2 the water magnetization exhibits decay times too close from the oil magnetization. In consequence, the total system response exhibits a single, largely spread T_2 peak. We conclude that the criterion of reservoir quality derived in case C3H4 (two distinct T_2 peaks at long TE) is only valid in the presence of micro-porosity.

Hypothesis 3 (C4H3): Unimodal pore structure, partially filled with light oil. The main objective of this simulation case is explore a difference with case C4H2 where, despite the same effective pore size left for water (same peak relaxation at $TE=0.25$ ms), the actual pore size and structure are completely different. It is observed that the unimodal pore structure exhibits a response at long echo times. Consequently, a more pronounced difference arises between the oil peak and the water peak. However, as shown by case C3H4, the pattern is not unique.

Hypothesis 4 (C4H4): Bimodal, hydraulically coupled pore structure partially saturated with light oil. As discussed earlier, a 100-ms T_2 peak can be the result of hydraulic coupling between micro- and macro-pores. In the presence of both irreducible water and partial light oil saturation, when TE increases this coupled system exhibits an important transfer of magnetization toward short relaxation times. Transfer of magnetization is such that the presence of oil is only evidenced by a tail toward long T_2 of the total response at high values of TE .

Hypothesis 5 (C4H5): This case is constructed with the same uncoupled pore configuration as in C4H1, except for the partial saturation of the macro-porous region with light oil. By direct comparison with the simulation results obtained for case C4H4, we remark that, in similarity with cases C3H2/C3H3, the coupled case C4H4 is not more amenable to phase discrimination than the uncoupled case C4H5. However, this fundamental difference, based on the transfer of the macro-porous borne magnetization to shorter values of T_2 , suggests a

practical way to discriminate between the coupled or uncoupled nature of the pore structure.

Conclusions

A novel numerical algorithm was developed to simulate NMR measurements acquired in the presence of a constant magnetic field gradient. The geometrical and petrophysical model of rock formations included dual fluid saturations and dual porosity structures. We applied this simulation algorithm to a number of examples of porous media systems and successfully reproduced actual NMR measurements acquired in complex pore geometries exhibiting hydraulic coupling among primary and secondary pores.

The algorithm was used to assess the influence of multi inter-echo times in the detection and quantification of partial oil- and water-phase saturations in carbonate rocks. We focused our simulation and interpretation work to cases where a sufficient contrast existed between the molecular diffusivity of the saturating fluids. It was found that presence of full light oil saturation in the macro-pores together with water-filled micro-porosity was evidenced by a clear distinction of the corresponding T_2 peaks at long values of TE . We also found that the presence of diffusive coupling can help to discriminate oil from water. It was observed that it is not always the best choice to use long values of TE for oil identification and saturation estimation in coupled pore systems. Such a standard discrimination method applied to coupled structures may not be suitable for the interpretation of uncoupled pore structures. However, the difference in the NMR response from these two pore systems (coupled and uncoupled) constitutes by itself a hint to the geometrical properties of the pore structure.

Nomenclature

Δt	= duration associated with a First-Passage-Time technique step (ms)
ϵ	= unitary displacement of a step of conventional random-walk technique applied near the pore wall (nm)
γ	= proton gyromagnetic ratio = $2\pi \cdot 4258$ rad/(G.s)
η	= fluid viscosity (cp)
ρ	= surface relaxivity ($\mu\text{m/s}$)
ϕ	= proton spin phase (rad)
D	= effective diffusivity (m^2/s)
D_o	= bulk diffusivity (m^2/s)
G	= strength of external magnetic field gradient applied by NMR tool (G/cm)
p	= probability of proton magnetization decay at fluid surface boundary
R	= macroscopic displacement of a step of First-Passage-Time technique (nm)
S/V	= surface-to-volume ratio of pore (μm^{-1})
S_w	= water saturation
S_{wir}	= irreducible water saturation
t	= time (ms)
T_2	= total transverse relaxation (ms)
T_{2B}	= bulk transverse relaxation (ms)
T_{2D}	= diffusive transverse relaxation (ms)

T_{2S} = surface transverse relaxation (ms)

TE = inter-echo time of the NMR tool (ms)

Acknowledgements

We express our gratitude to Baker Atlas for partially funding this work through an internship position offered to ET during the Summer 2001. We also like to thank Drs. Gigi Zhang and Carl Edwards for useful discussions. Partial support is acknowledged from the Center of Excellence in Formation Evaluation of The University of Texas at Austin, and from the American Chemical Society under grant ACS PRF#37519-AC9. The Center of Excellence in Formation Evaluation is an industry research consortium jointly sponsored by Baker Atlas, Halliburton, Schlumberger, and Anadarko.

References

- 1 Chen, S., Georgi, D. T., Olima, O., Gamin, H., and Minetto, J. C.: "Estimation of hydrocarbon viscosity with multiple-TE, dual-TW MRIL logs", SPE Reservoir evaluation and engineering (December 2000), 3, 6.
- 2 Akkurt, R., *et al.*: "Enhanced diffusion: Expand the range of NMR direct hydrocarbon-typing applications", paper GG presented at the 1998 SPWLA Annual Logging Symposium, Keystone, Colorado, May 26-29.
- 3 Looyestijn, W. J.: "Determination of oil saturation from diffusion NMR logs", paper SS presented at the 1996 SPWLA Annual Logging Symposium, New Orleans, Louisiana, June 16-19.
- 4 Ramakrisnan, T. S., Schwartz, L. M., Fordham, E. J., Kenyon, W. E., and Wilkinson, D. J.: "Forward models for nuclear magnetic resonance in carbonate rocks", The Log Analyst (July-August 1999), 40, 4.
- 5 Toumelin, E.: Monte Carlo simulations of NMR measurements in carbonate rocks under a constant magnetic field gradient, Master's thesis, The University of Texas at Austin, Austin, TX (2002).
- 6 Zheng, L. H., and Chiew, Y. C.: "Computer simulation of diffusion-controlled reactions in dispersions of spherical sinks", Journal of Chemical Physics (1989), 90, 1.
- 7 Coates, G. R., Vinegar, H. J., Tutunjian, P. N., and Gardner, J. S.: "Restrictive diffusion from uniform gradient NMR well logging", SPE 26472, 1993.
- 8 Gudjartsson, H., and Patz, S.: "NMR diffusion simulation based on conditional random walk", IEEE Transactions on medical imaging (December 1995), 14, 4.
- 9 Bergman, D. J., Dunn, K.-J., Schwartz, L. M., and Mitra, P. P.: "Self-diffusion in a periodic porous medium: a comparison of different approaches", Physical Review E (April 1995) 51, 4.
- 10 Bergman, D. J., Dunn, K.-J., Schwartz, L. M., and Mitra, P. P.: "Self-diffusion in a periodic porous medium: a comparison of different approaches", Physical Review E (April 1995) 51, 4.
- 11 Chen, S., Fang, S., Georgi, D. T., Salyer, J., and Shorey, D.: "Optimization of NMR Logging Acquisition and Processing", paper SPE 56766 presented at the 1999 SPE Annual Technical Conference and Exhibition, Houston, Texas, October 3-6.
- 12 Toumelin, E., Torres-Verdin, C., Chen, S., Fischer, D. M.: "Analysis of NMR diffusion coupling effects in two-phase carbonate rocks: comparison of measurements with Monte Carlo simulations", paper JJJ presented at the 2002 SPWLA Annual Logging Symposium, Oiso, Japan, June 2-5.

Fluid	η (cp) at 370 K	D_o ($\mu\text{m}^2/\text{ms}$)	T_{2B} (ms)
Brine (water)	1	2.5	3700
Light oil	1.5	1	1000
Medium oil	6	0.27	250

Table 1: Summary of bulk fluid properties used for the simulation of NMR fluid saturation effects

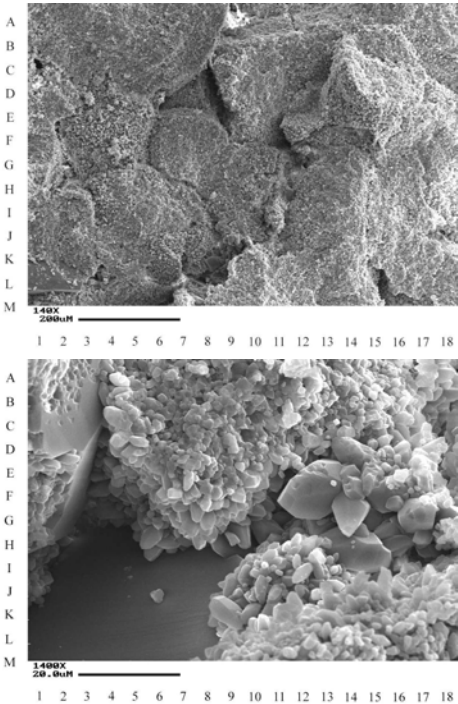


Fig. 1: SEM images of carbonate rock at 140x (top panel) and 1400x (bottom panel) magnifications. A dual porosity structure is clearly evidenced at these two scales: from a diagenetic viewpoint, the grains seen on the top panel were micritized into micro-grains as evidenced on the bottom panel. Therefore, the solid matrix is formed by micro-grains agglomerated into packs or pseudo (micro-porous) macro-grains.

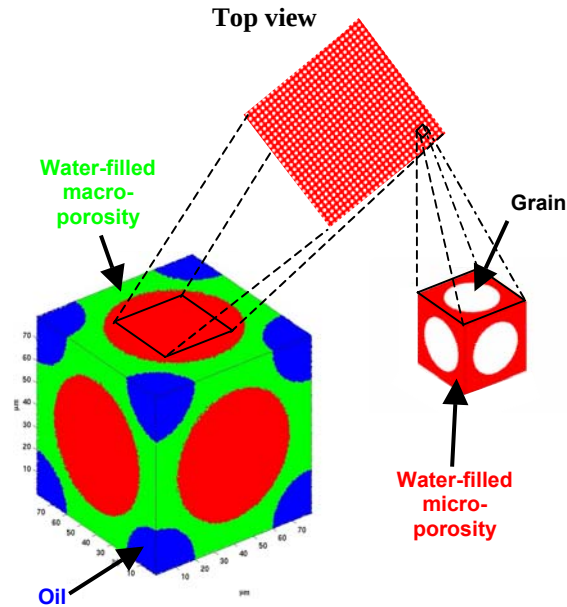


Fig. 2: Three-dimensional representation of the distribution of fluids and rock matrix at the spatial resolution of both macro- and micro-cells. Gas blobs are centered in the macro-pores. Grain compaction is simulated by assigning sphere diameters larger than the concentric cube sizes.

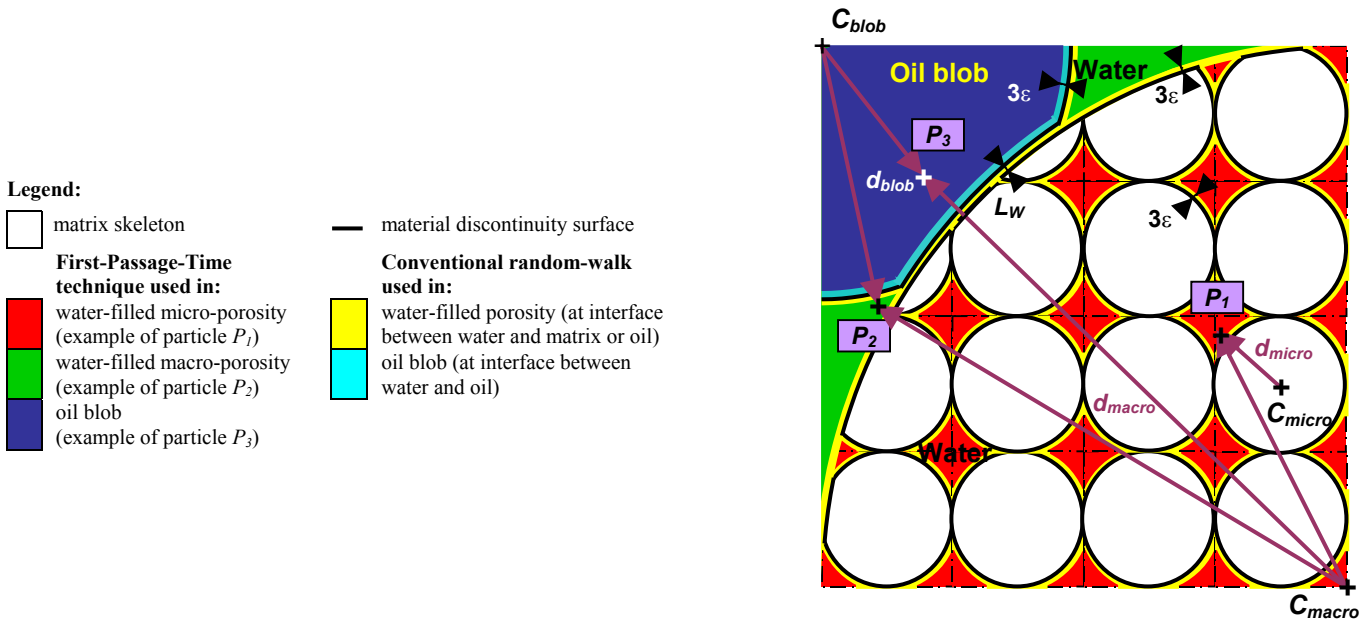


Fig. 3: Fraction of a cross-section taken through a pack period. The graph illustrates the criterion used to select a specific random-walk technique according to the location of the particles at the onset of the walk step. A nominal distance of 3ϵ is taken as the safety thickness within which the First-Passage-Time technique reverses to a conventional random walk. A wetting layer of thickness L_w is also included in order to constrain the rock wettability at the surface of the micro-porous grain (grain pack).

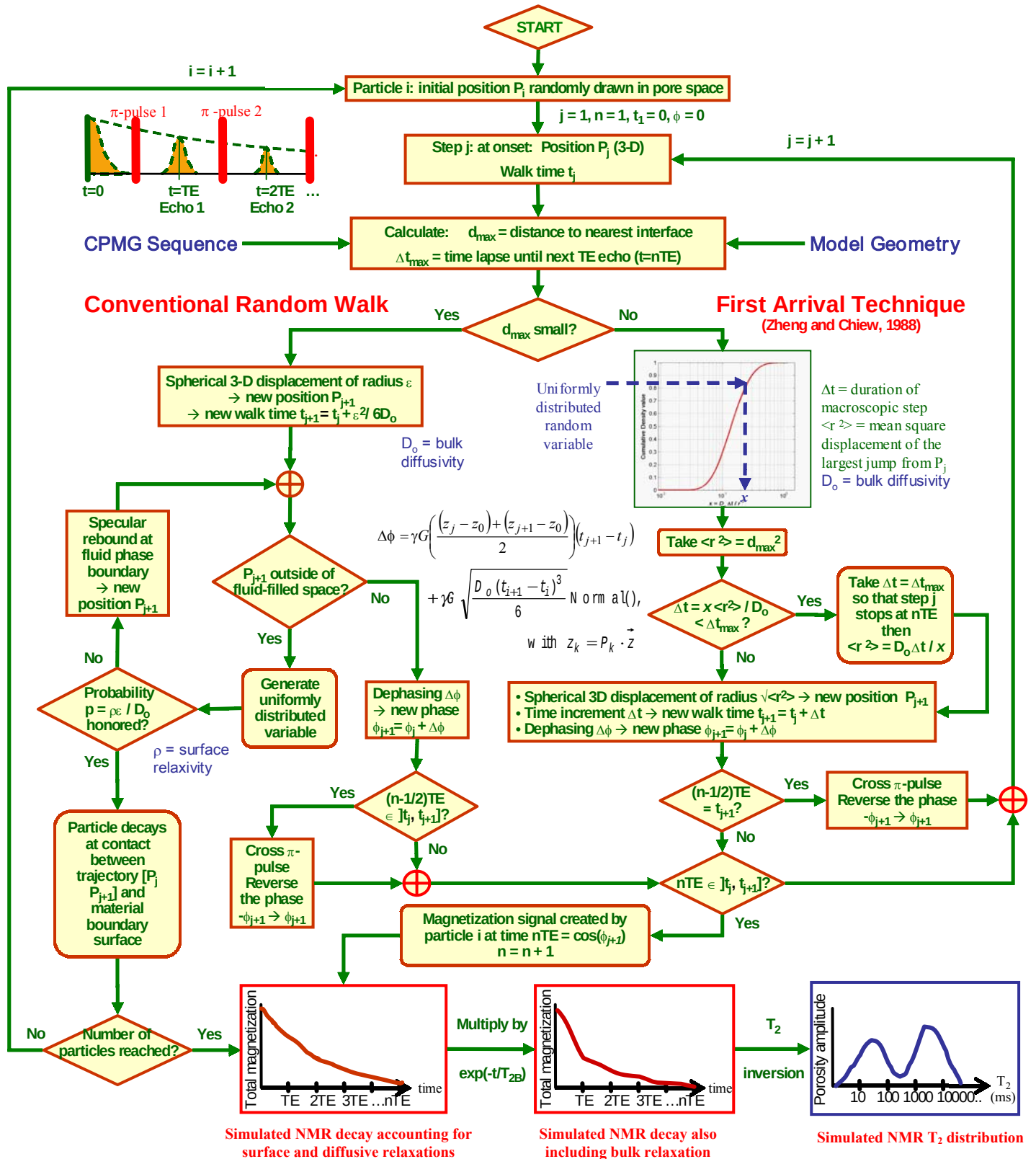


Fig. 4: Flowchart of the NMR simulation algorithm used for each fluid phase. In the descriptions, n is the echo index of the CPMG sampling, i is the particle index, j is the step index along particle i 's walk, P_i is the 3D position of particle i at the onset of step j , t_i is the walk time of particle i at the onset of step j , and ϕ_i is the spin phase of particle i at the onset of step j .

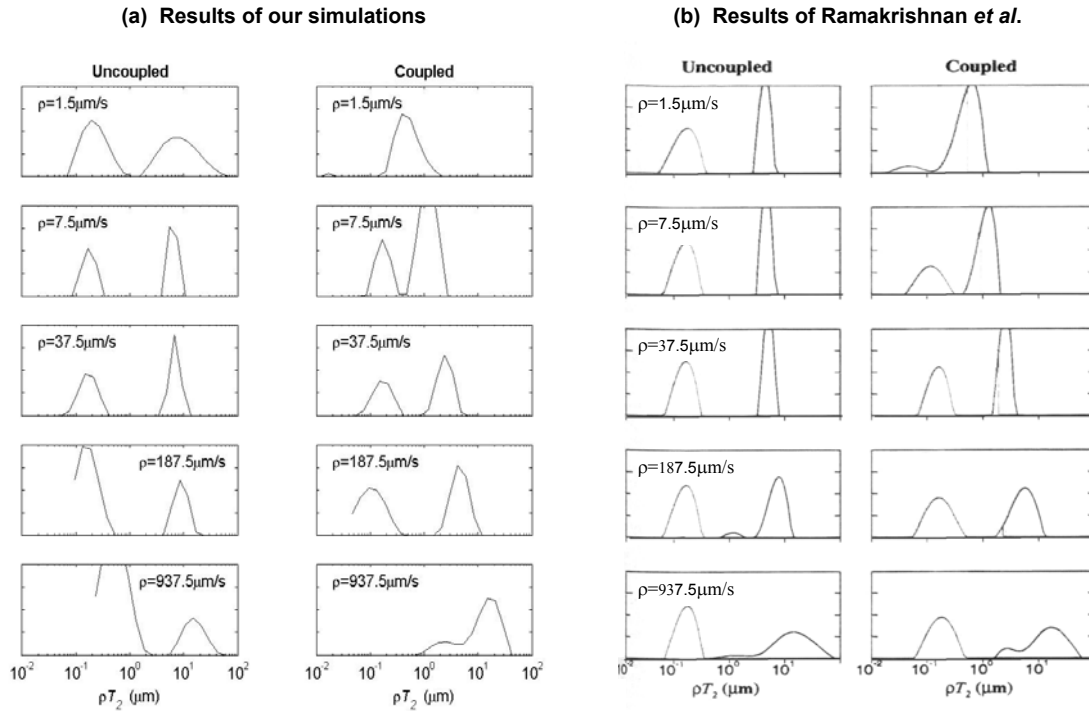


Fig. 5: Comparison of simulations of NMR data transformed into equivalent pore-size distributions. (a) Simulations performed with our simulation program, and (b) simulations reported by Ramakrishnan *et al.*⁴, using a single water phase and different values of surface relaxivity ρ . Equivalent pore-size distributions were obtained by multiplying ρ and simulated T_2 distributions, and exhibit the importance of ρ in NMR diffusion coupling. Although our algorithm is unable to detect low T_2 values in the theoretical case of high ρ values, the agreement between the two sets of simulations is remarkable.

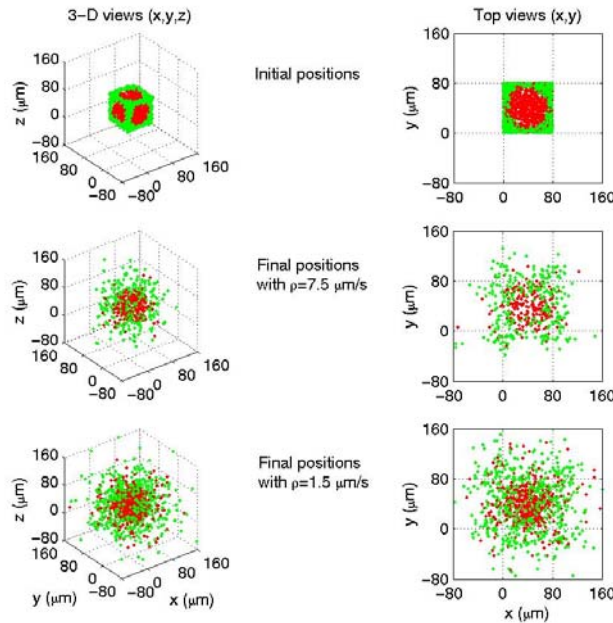


Fig. 6: Illustration of the spatial dispersion of fluid particles as a function of surface relaxivity ρ . Initially, within a period of the dual pack of spheres, dark red dots indicate locations of particles borne by the micro-pore space, whereas light green dots indicate locations of particles borne by the macro-pore space. Final particle positions are either the locations of the particles at their decay, or else current locations of non-decayed particles decayed after a 1-second lapse.

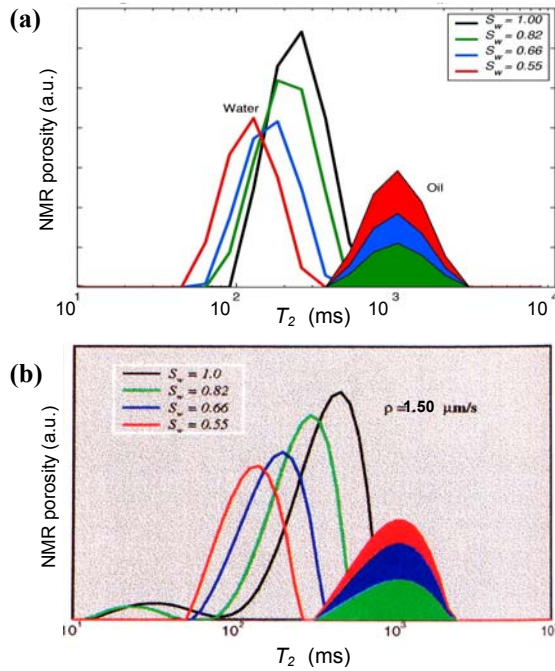


Fig. 7: Effect of non-wetting oil saturations on the NMR response of hydraulically coupled systems. Simulation results of (a) our algorithm, and (b) Ramakrishnan et al.⁴, obtained for similar coupled geometries. Colored peaks stand for the magnetization within the oil phase, whereas colored lines limit the T_2 peaks within wetting water.

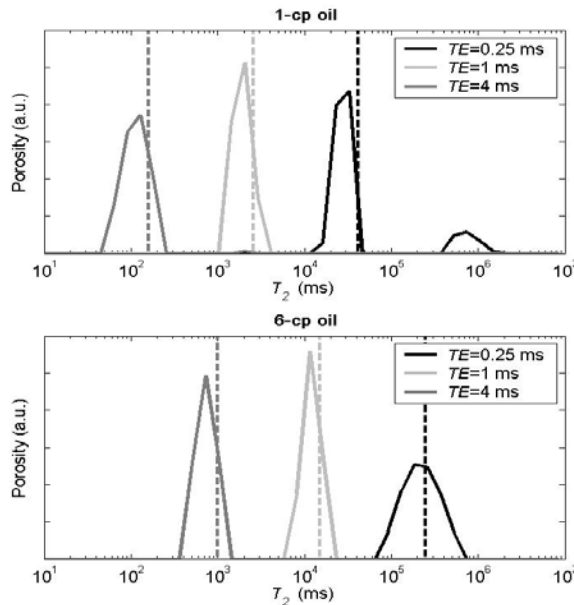


Fig. 8: Comparison of analytical T_{2D} relaxations (dash) and simulated T_2 distributions (solid) of 1-cp and 6-cp oil grades in unbounded diffusion. The simulation results are considered accounting for no bulk relaxation [before multiplying the output decay by the function $\exp(-t/T_{2B})$]. T_{2D} values are calculated from equation (3), for $TE=0.25, 1$, and 4 ms, i.e., 160, 2600, and 41500 ms for light oil (1 cp), and 1000, 15000, and 248000 ms for medium oil (6 cp), respectively.

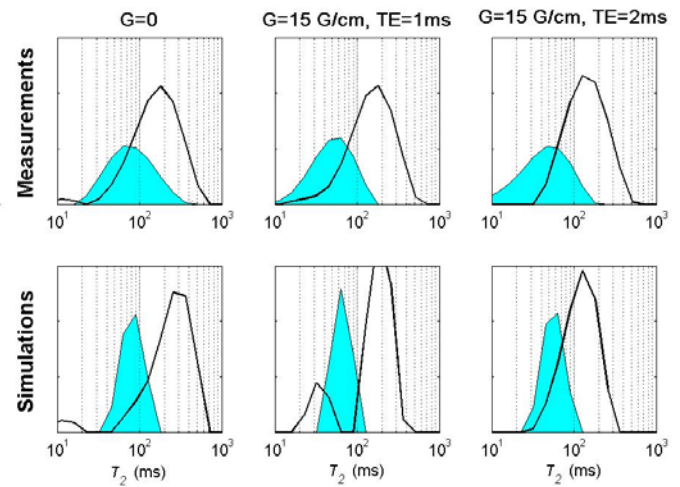


Fig. 9: Example of comparison of measured and simulated T_2 distributions in the absence of a magnetic field gradient¹¹ ($G=0$), and then with ($G=15$ G/cm), for echo times of 1 and 2 ms. Blue (respectively, in color) distributions correspond to measurements and simulations performed at irreducible water saturation (respectively, 100% brine saturation). A very good agreement exists in terms of distribution spread for 100% brine saturation and T_2 peak value for both saturations, especially considering that the exclusively bimodal distribution of the model responsible for the simulated T_2 peaks sharper than in the measured T_2 distribution..

CASE 1: unimodal distribution, high T_2

Hypothesis 1: unimodal pore structure, water only

Hypothesis 2: unimodal pore structure, partially filled with light oil

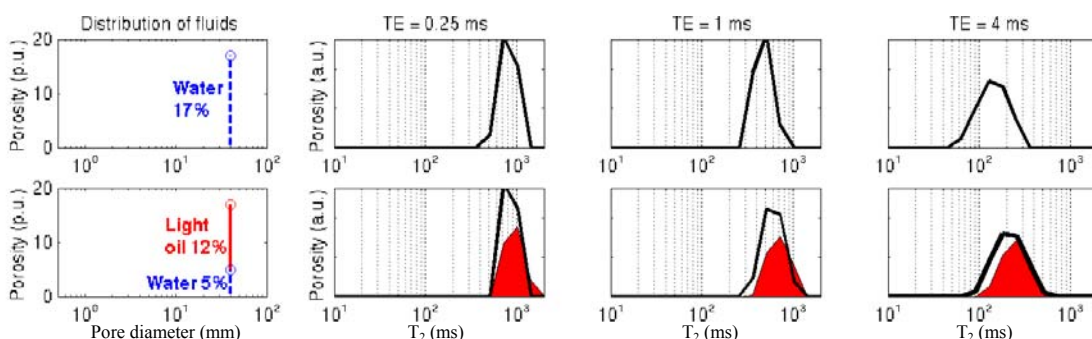


Fig. 10: CASE 1 simulation results. Each row states a hypothesis, symbolizes it in terms of a fluid distribution within the pore system, and presents the results of the corresponding simulations for three values of inter-echo time. The first column of graphs summarizes the distribution of fluids in the pore space. Dashed blue (resp. plain red) stems represent the porosity fractions of water (resp. oil) in the pore space characterized by the nominal pore sizes described in the horizontal axis of the plot. Fluid content is indicated in porosity units. The remaining columns describe the T_2 distributions resulting from the simulations with a 15 G/cm gradient, and TE values of 0.25, 1, and 4 ms. Red-filled T_2 distributions correspond to the magnetization within oil only, whereas unfilled T_2 distributions represent the magnetization within the whole pore system (i.e., including all fluid phases). Total magnetization can be understood as the one acquired with the NMR tool.

CASE 2: unimodal distribution, average T_2

Hypothesis 1: unimodal pore structure, water only

Hypothesis 2: unimodal pore structure, partially filled with medium-grade oil

Hypothesis 3: coupled pore structure, water only

Hypothesis 4: coupled pore structure, macropores partially filled with medium-grade oil

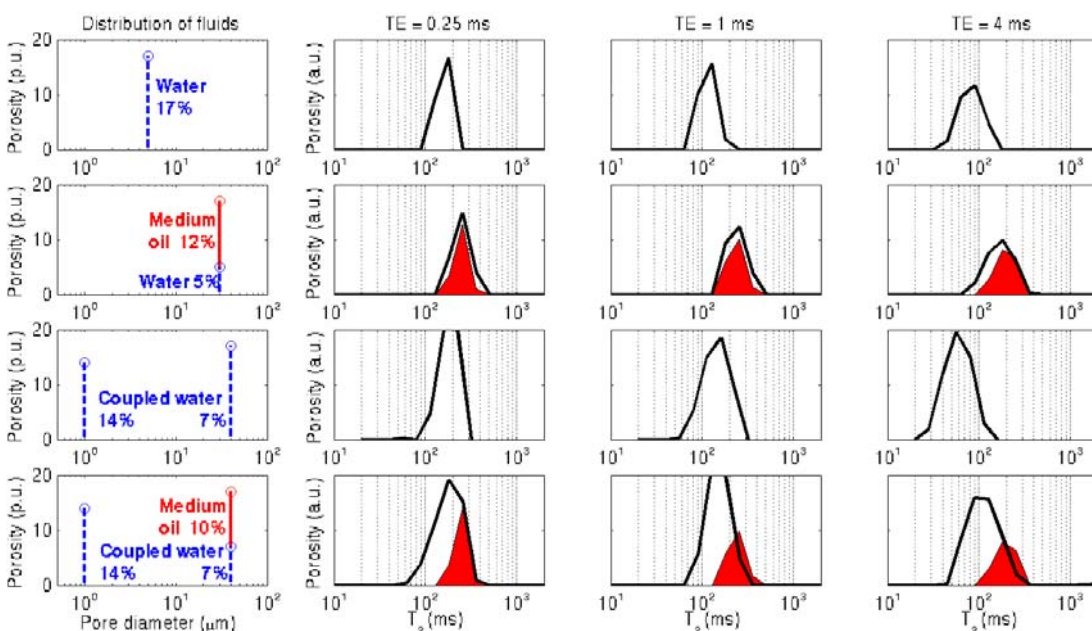


Fig. 11: CASE 2 simulation results. Refer to the caption of Fig. 10 for details on the plotting conventions. Red-filled T_2 distributions correspond to the magnetization within oil only, whereas unfilled distributions represent the magnetization within the whole pore system (i.e., also including water).

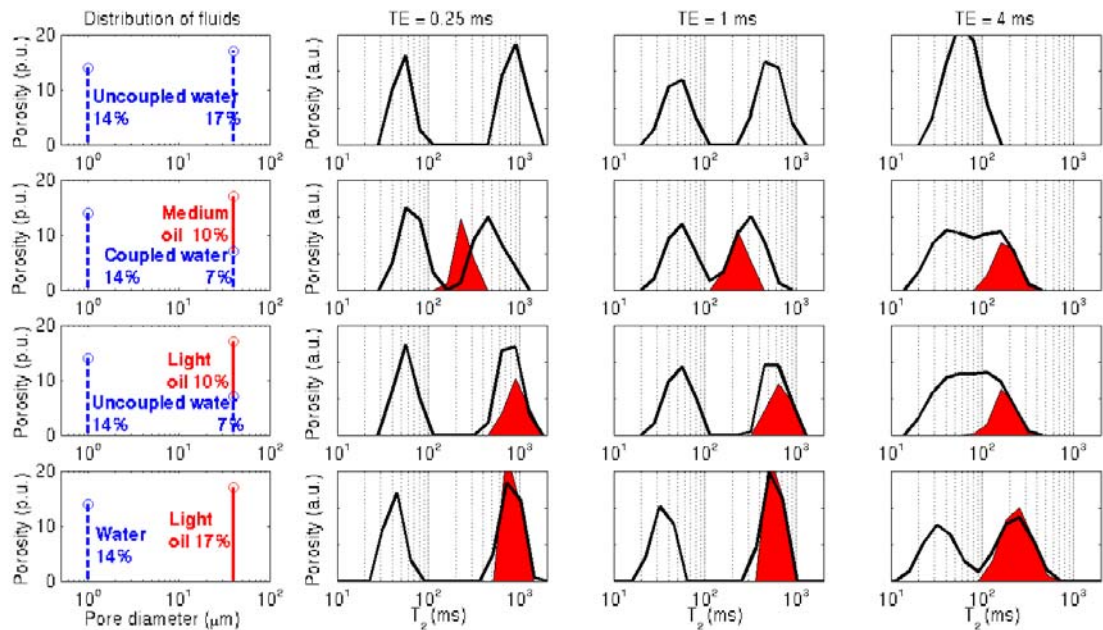
**CASE 3: bimodal
distribution, low/high
 T_2** **Hypothesis 1:** uncoupled
pore structure, water only**Hypothesis 2:** uncoupled
pore structure, macro-
pores partially filled with
medium oil**Hypothesis 3:** uncoupled
pore structure, macro-
pores partially filled with
light oil**Hypothesis 4:** coupled or
uncoupled pore structure,
macro-pores totally filled
with light oil

Fig. 12: CASE 3 simulation results. Refer to the caption of Fig. 10 for details on the plotting conventions. Red-filled T_2 distributions correspond to the oil only, whereas unfilled distributions correspond to the whole pore system (also including water).

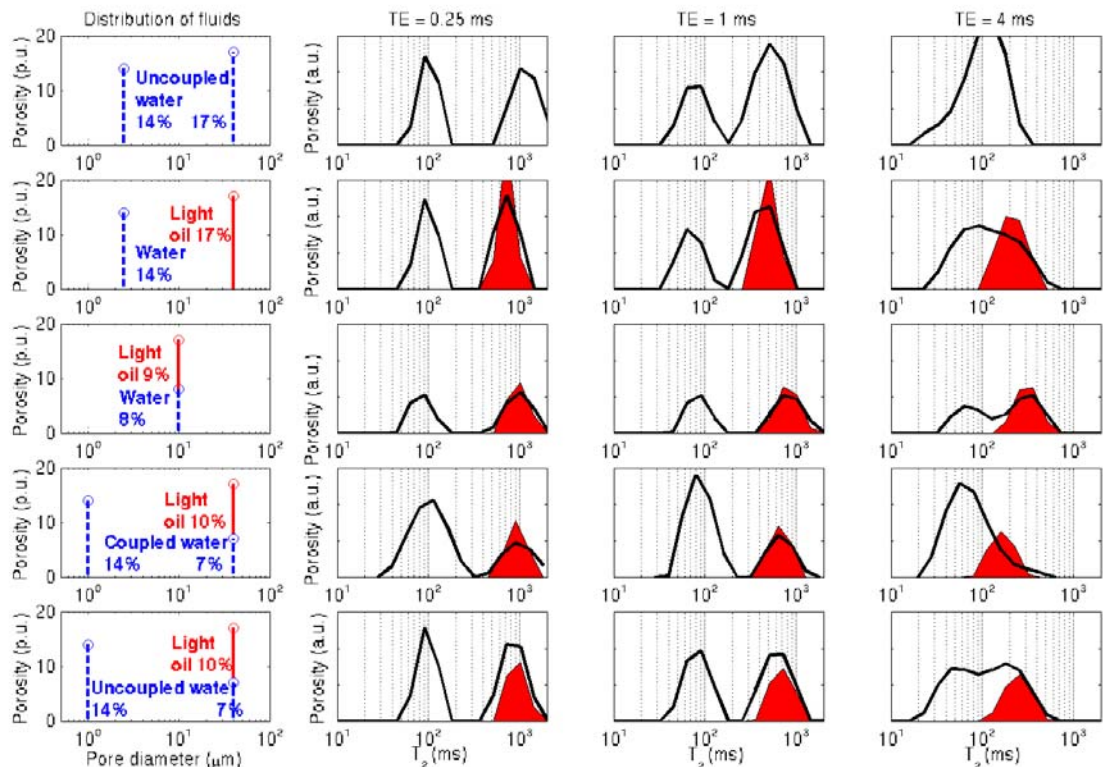
**CASE 4: bimodal
distribution,
average/high T_2** **Hypothesis 1:** uncoupled
pore structure, water only**Hypothesis 2:** coupled or
uncoupled pore structure,
macro-pores completely
filled with light oil**Hypothesis 3:** unimodal
pore structure, partially
filled with light oil**Hypothesis 4:** coupled
pore structure, macro-
pores partially filled with
light oil**Hypothesis 5:** uncoupled
pore structure, macro-
pores partially filled with
light oil

Fig. 13: CASE 4 simulation results. Refer to the caption of Fig. 10 for additional details on the plotting conventions. Red-filled T_2 distributions correspond to the oil only, whereas unfilled distributions correspond to the whole pore system (also including water).